

Image Generation and Editing APIs

From Random Pixels to Controlled Production Pipelines

THE 10/90 AXIOM IN COMPUTER VISION

«The 10/90 Axiom: The model API call is only 10% of the system. 90% of production reality is masks, geometric preservation, S3/GCS object storage, Base64 decoding, metadata audit trails, and moderation review gates.»

01 • PRIMITIVES

Visual Primitives

Text-to-Image, Inpainting, Outpainting, masks, and decoupled reference signals.

02 • CONTROL

Contractual Control

Prompt anatomy as an engineering spec: sacred attributes and acceptance criteria.

03 • PIPELINE

Production Pipeline

Metadata audit trails, Base64 decoding, Blast Radius, and Human-in-the-Loop gates.

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ENTERPRISE DELIVERY

Architecture & Systems Leadership

15+ years in enterprise software, distributed leadership, and embedding AI assistants into engineering workflows.

HANDS-ON AI R&D

Architect of Creator Tools (Pet-Project)

Creator and architect of Creator Tools — a suite of AI tools for YouTube creators designed to save production time, streamline content workflows, and increase earnings.

PRODUCTION CASE

EPAM CodeMie @ Dawn Foods

35% SDLC acceleration across feature development, unit testing, and architectural documentation.

OFFICIAL CONTACT

Homework Submissions

GitHub collaboration and code reviews: ihar_rubanovich@epam.com

Curriculum Roadmap: Where We Are

16 lessons across 6 modules: from text BPE tokens to multimodality and autonomous agents

MODULE 1 • L01-L03

Foundations & Core API

BPE tokens, prompt engineering, context windows, Python API calls, streaming, and structured JSON.

✓ COMPLETED

YOU ARE HERE • MODULE 2

Multimodality

- L04: Image APIs & Inpainting
- L05: Video Generation (Runway, Kling, Sora)
- L06: Speech & Audio (ElevenLabs, Whisper)

▶ CURRENT PHASE

MODULE 3 • L07-L09

Developer Tooling

GitHub Copilot, Claude Artifacts, Advanced Data Analysis, and automated code generation.

⌚ UPCOMING

MODULE 4 • L10

Local AI & Open Weights

LM Studio, Ollama, weight quantization, local inference, and cloud deployment on RunPod.

⌚ UPCOMING

MODULE 5 • L11-L13

Agents & MCP Protocol

Function Calling, building first autonomous agents, Model Context Protocol architecture.

⌚ UPCOMING

MODULE 6 • L14-L16

Production & Agentic IDEs

Streamlit rapid prototyping, safety/guardrails, Cursor, Windsurf, and Claude Code CLI.

⌚ UPCOMING

GitHub Workflow & Artifact Structure

Standardized homework submission: lesson-04 branch and reproducible artifact package

REPOSITORY TREE • L04

```
genai-homeworks/
├─ L01/
├─ L02/
├─ L03/
└─ L04/
  ├─ brief.md # Visual Generation Canvas
  ├─ references/ # Source references & masks
  ├─ prompt_v1.txt # Initial prompt
  ├─ prompt_v2.txt # Production prompt with sacred rules
  ├─ candidates/ # Generated candidate outputs
  ├─ qa.md # Defect audit scorecard
  └─ selected/ # Accepted asset + audit metadata
```

SUBMISSION PROTOCOL

lesson-04 Branch & Pull Request

All changes committed to feature branch. Open PR to main, add ihar_rubanovich@epam.com as reviewer. Zero direct pushes to main.

GOVERNANCE & SAFETY

Data Privacy & Brand Governance

No real faces without consent, confidential blueprints, or unreleased assets. Use open synthetic references.

The Trap: Pretty ≠ Usable

Why an aesthetically flawless image can become a silent catastrophe in enterprise systems

ENTERTAINMENT VS PRODUCTION

The Illusion of Perfection

In LLMs, failures are obvious upon reading: broken syntax or factual errors.

In visual generation, **an unusable image can look cinematic and gorgeous**. The human eye is seduced by lighting and fails to spot critical geometric drift.

5 SILENT PRODUCTION FAILURES

- 01 Product geometry drift: Handle shifted by 15%, ceramic curvature morphed — customer receives a different physical item.
- 02 Brand logo hallucination: Official trademark mutates into distorted pseudo-text.
- 03 Factual misrepresentation: Hallucinating non-existent features (e.g. EU wall outlet in US campaign).
- 04 Copyright & privacy risks: Accidental replication of protected artist styles or unconsented likenesses.
- 05 Zero reproducibility: Unrepeatable outputs that cannot be systematized in production pipelines.

Modality Shift: Diffusion Physics & Latent Space

Why images are not discrete BPE tokens, and how iterative denoising operates in latent space

Text LLMs:
Discrete BPE tokens \$w_1 \to w_2 \to w_3\$, autoregression. |

Diffusion & DiT:
Continuous tensors, VAE latent space, and iterative Denoising.

1. VAE LATENT SPACE

Latent Space Compression

Millions of RGB pixels compressed by Variational Autoencoder (VAE) by 8x-16x into compact latent space z for efficient inference.

2. FORWARD & REVERSE DENOISING

Iterative Denoising

Model trained to reverse Gaussian noise over 20-50 steps. UNet or DiT predicts noise increments at each discrete step.

3. CONDITIONING & SEED

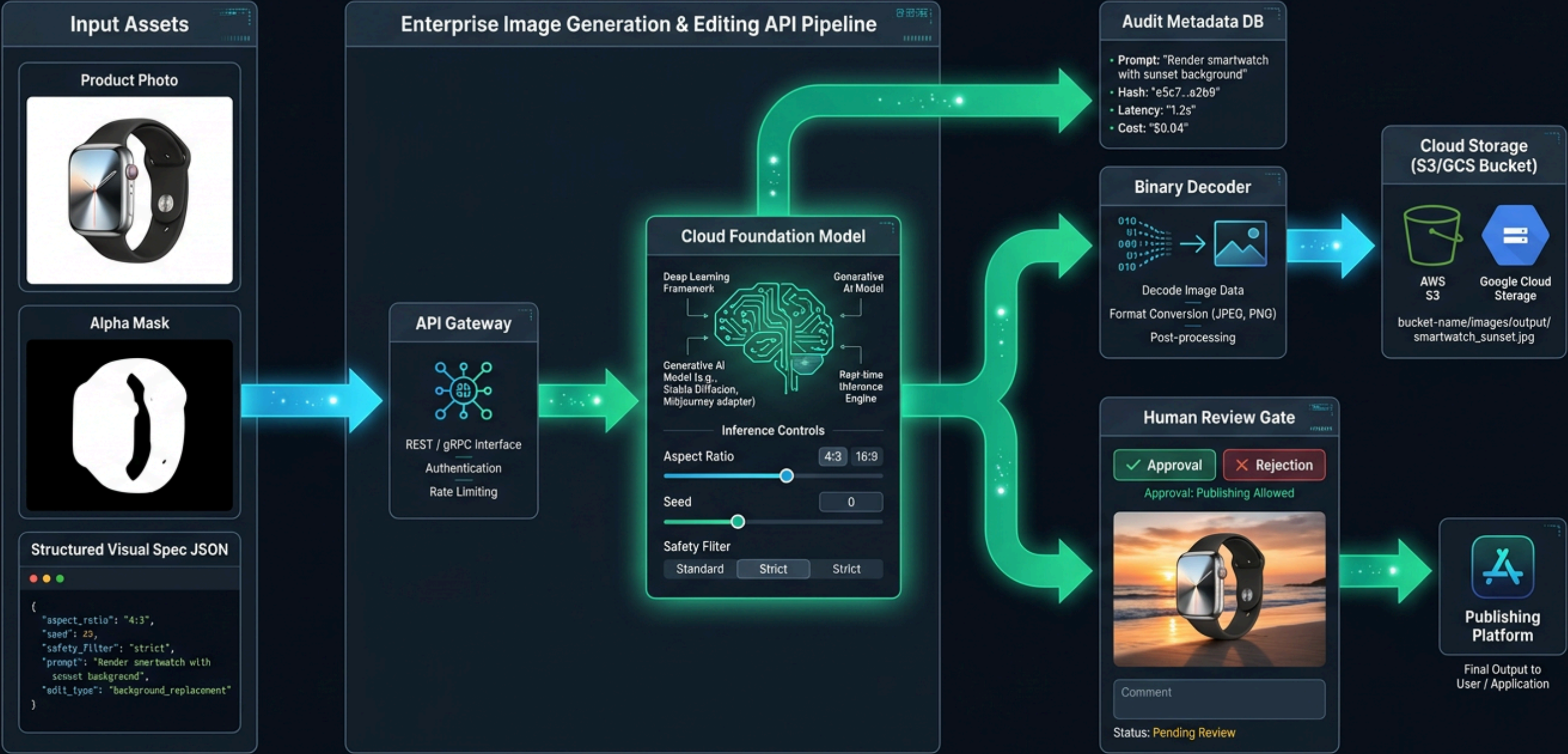
Conditioning & Seed

CLIP/T5 text encoder directs denoising via Cross-Attention. Seed pins initial Gaussian distribution for reproducible outputs.

Mental Model: Generation as a Pipeline

The model is only one node; predictability is engineered in the pipeline before and after inference

C4 Architectural Infographic Diagram: Enterprise Image Generation and Editing API Pipeline



The 6 Visual Primitives of Image AI

The foundational operations combined to solve all commercial visual engineering workflows

01 • TEXT-TO-IMAGE

Synthesis From Scratch

Synthesizing pixels from text alone. Best for concept ideation and storyboard backdrops.

02 • IMAGE-TO-IMAGE

Reference Transformation

Conditioning on source image plus prompt. Preserves spatial composition while restyling.

03 • INPAINTING

Masked Region Editing

Modifying pixels strictly within mask boundary. Swapping product background while freezing product.

04 • OUTPAINTING

Canvas Outpainting

Generating scene beyond original canvas borders. Expanding 1:1 square photo to 16:9 banner.

05 • REFERENCE SIGNALS

Multimodal Conditioning

Pinning product identity (IP-Adapter), aesthetics (Style Reference), or pose skeletons (ControlNet).

06 • NEGATIVE CONSTRAINTS

Negative Constraints

Deterministic exclusion: stripping unwanted text, watermarks, duplicate objects, and anatomical glitches.

Prompt Anatomy: An Engineering Brief, Not a Poem

Deconstructing visual prompts into explicit parameters instead of decorative adjectives

SUBJECT • HERO OBJECT

Focal Entity

Single hero object, physical material, glaze texture, surface reflectance. Zero ambiguity.

COMPOSITION • FRAMING

Framing & Camera

Eye-level product macro, shallow depth of field, clean negative space reserved for typography.

ENVIRONMENT • LIGHTING

Scene & Lighting

Surface material (travertine), lateral natural light from left, realistic contact shadows.

STYLE • AESTHETIC

Aesthetic Language

Commercial editorial photography. Explicit prohibition of 3D renders or illustrations.

SACRED ATTRIBUTES • TABOO

Sacred Attributes (Taboo)

Most critical line: mug geometry, handle curvature, and logo must remain untouched.

ACCEPTANCE • CRITERIA

Acceptance Criteria

Checklist: catalog contour match, zero unwanted text, realistic lighting, moderation pass.

Before vs After: Weak Prompt vs Production Spec

How the exact same task executes as an unpredictable gamble vs a deterministic contract

WEAK PROMPT (GAMBLE)

"Make a beautiful ad image for this mug in a cozy studio."

Typical Failure:

- Model takes 99% creative freedom.
- Handle geometry morphs, glaze drifts.
- Unsolicited props and duplicate mugs hallucinated.
- *Unusable for commercial catalog pipelines.*

PRODUCTION SPEC (PIPELINE)

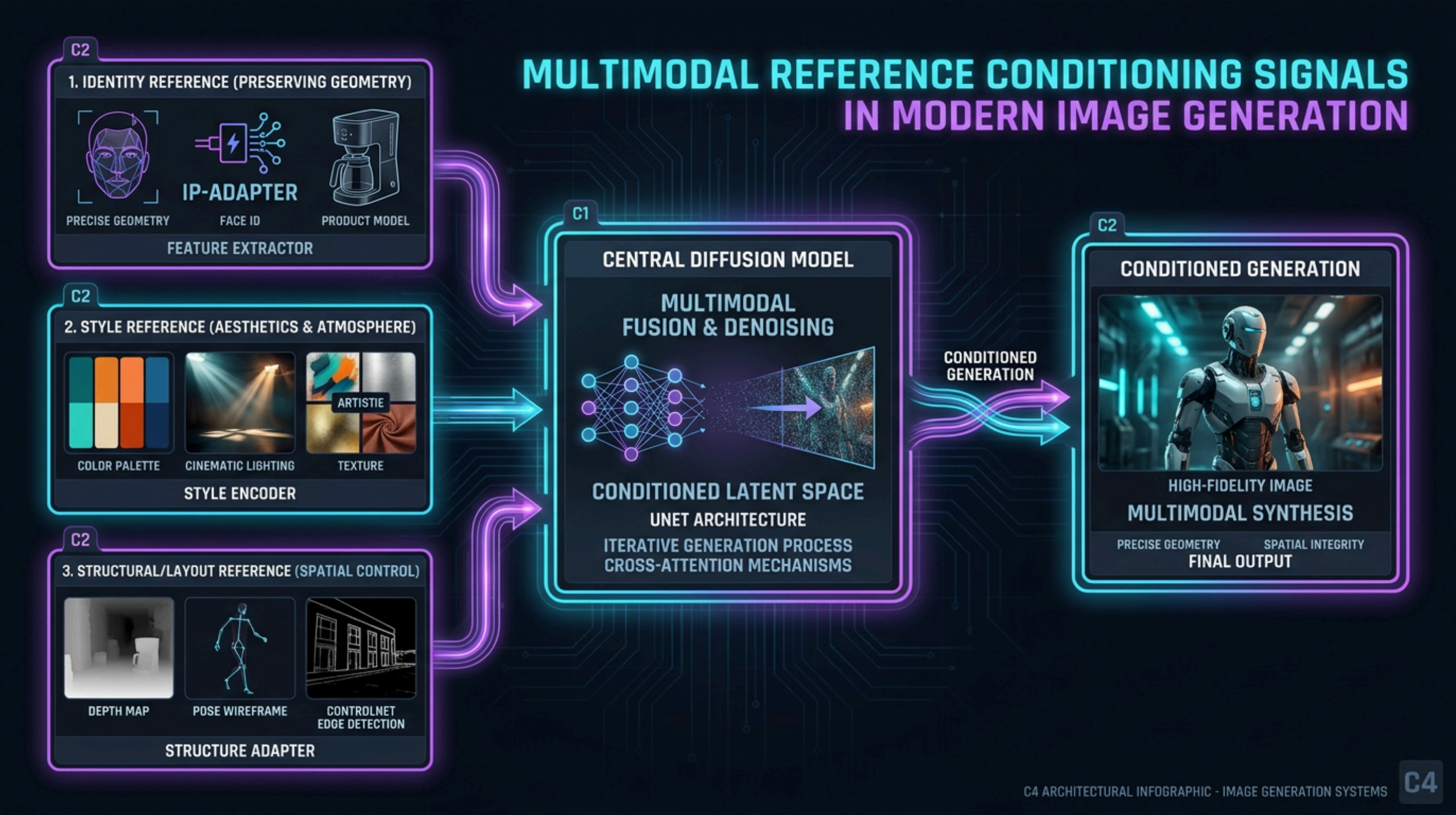
"Using input_product.png as protected reference, preserve mug shape, speckled glaze, handle ergonomics, and proportions 100%. Replace only background and surface with pale travertine studio in soft natural morning light. Eye-level, 16:9 crop, 2K. No text, no hands, no extra mugs. Return 3 variants. Acceptance: zero geometry drift against catalog master."

Production Outcome:

- Isolated parameters parameterized via backend JSON.
- 100% physical geometry and brand preservation.
- Deterministic criteria for automated QA gates.

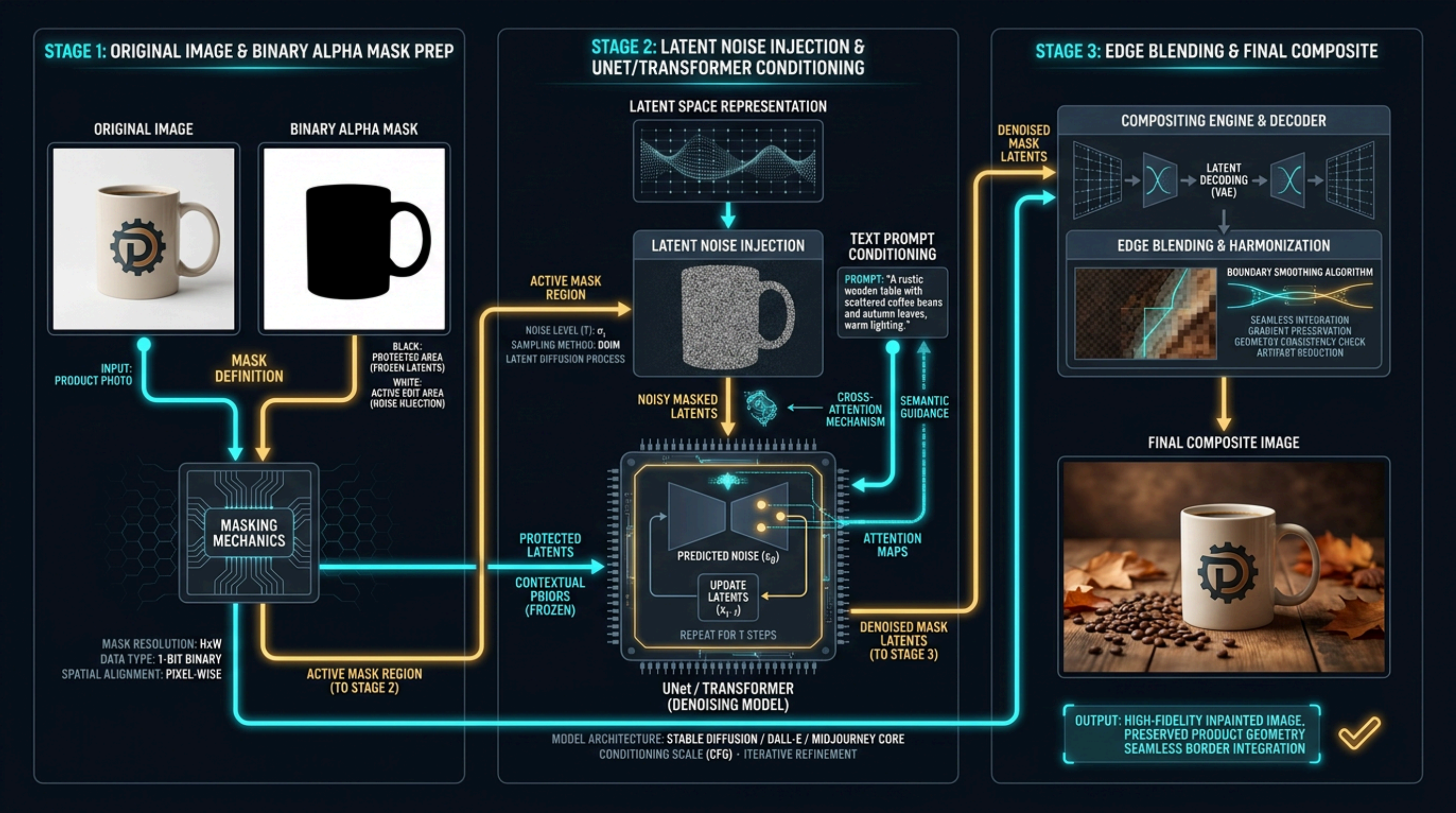
Reference Signals & Multimodal Control

Decoupling conditioning vectors: product identity (IP-Adapter), aesthetics (Style), and geometry (ControlNet)



Inpainting & Masking Mechanics

How binary alpha masks and edge feathering protect physical product geometry



API Lifecycle & Network Payloads

Traversing the wire: Synchronous Base64 payloads vs Asynchronous Job Queue Polling

PATTERN 1 • SYNCHRONOUS BASE64

Direct Byte Stream (2-5s)

Where used: Google Gemini API, OpenAI Images API.

- Client sends JSON with Base64-encoded image.
- HTTP socket remains open for 3-6 seconds.
- Server returns Base64 bytes directly in response payload.
- **Pros:** Immediate interactive conversational flow.
- **Cons:** Gateway timeout risk on batch 4K generation.

PATTERN 2 • ASYNC JOB QUEUE

Polling & Webhooks (30-60s)

Where used: Adobe Firefly Services, Midjourney, ComfyUI.

- Request immediately returns `HTTP 202 Accepted` with `job_id: 'img_98765'`.
- Backend polls status endpoint or handles webhook.
- Output downloaded via ephemeral signed S3 URL.
- **Pros:** Robust asynchronous decoupling.

Provider Landscape Without Hype

Comparing primary 2026 visual platforms by architecture, speed, and enterprise legal indemnity

OPENAI IMAGES & RESPONSES

GPT-Image-2 / DALL-E

High prompt comprehension, generation, and editing. Seamless integration as an agent tool via Responses API.

GOOGLE CLOUD / GEMINI

Gemini 3.1 Flash Image & Imagen 3/4

Native multimodal conversational context: iterative image editing via dialog refinement. Low-latency Base64 streaming.

ADOBE FIREFLY SERVICES

Firefly Image5 & Photoshop API v2

Enterprise benchmark: trained exclusively on licensed stock, full commercial indemnity, Photoshop PSD layer APIs.

OPEN SOURCE & LOCAL HOSTING

FLUX.1 / SDXL on RunPod

Total weight ownership, on-premise execution, zero data exfiltration, unrestricted ControlNet and LoRA fine-tuning.

Midpoint Break: 5 Minutes

Time for cognitive rest before diving into the 10/90 iceberg, Blast Radius, and live demo code

INTERACTIVE BREAK TIMER

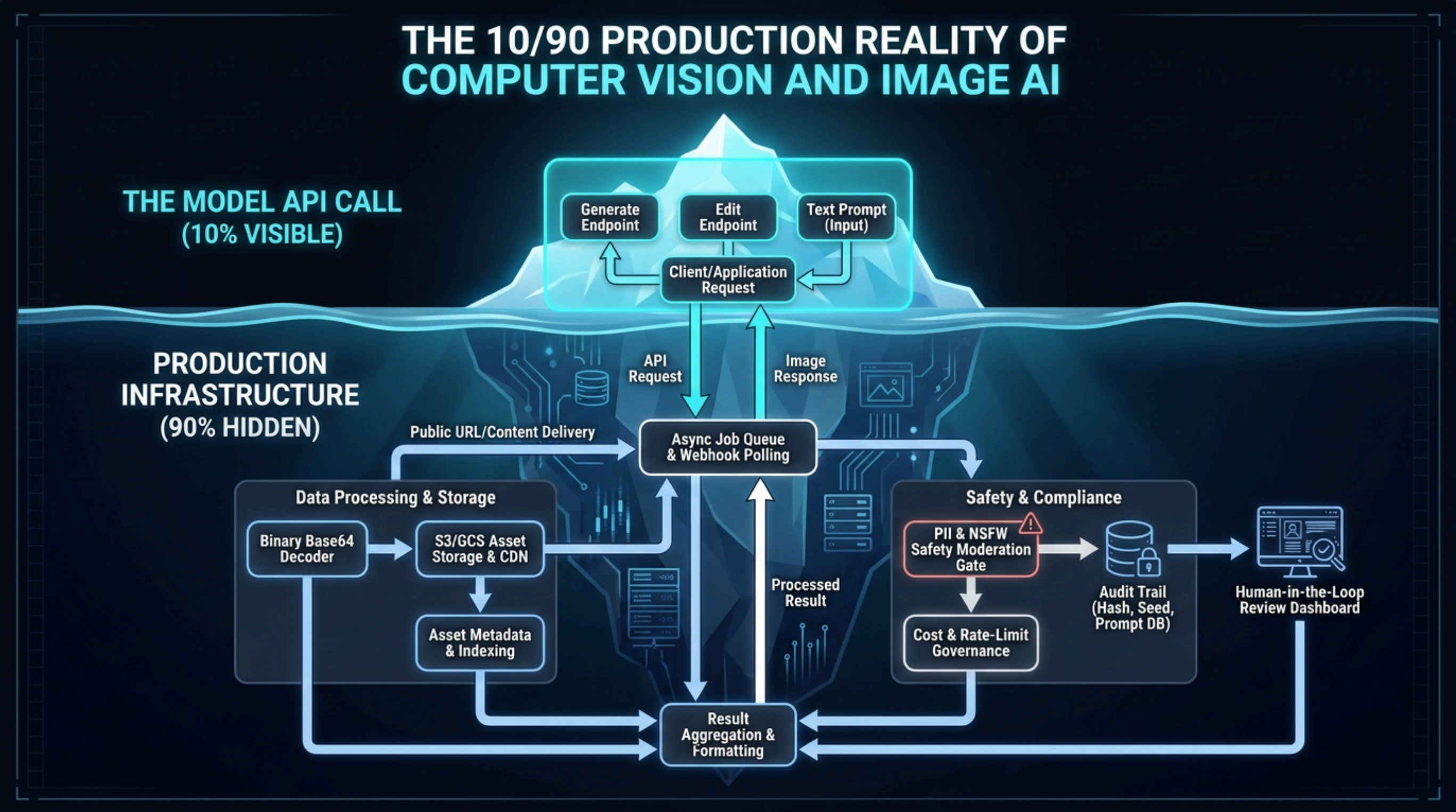
05:00

START 5 MIN RESET

Step away, hydrate, and stretch. Pure cognitive rest — no chat work discussions. In 5 minutes: the 10/90 iceberg, Blast Radius, live demo code, and homework.

The 10/90 Iceberg in Computer Vision

The 10% visible model API call vs 90% submerged production engineering infrastructure



Blast Radius Governance in Image AI

Traffic-light autonomy matrix: where AI is safe vs where autonomous execution is forbidden

● GREEN TIER • AUTONOMOUS

Zero Business Risk

Scenarios:

- Internal design moodboards.
- Synthetic profile avatars for QA load testing.
- Brainstorming exploratory sketches.

Autonomous execution permitted. Zero business blast radius.

● YELLOW TIER • HUMAN-IN-THE-LOOP

Moderate Operational Risk

Scenarios:

- Marketplace e-commerce backgrounds.
- Editorial blog graphics.
- YouTube video thumbnails.

Model acts as copilot; human editor holds final publishing sign-off.

● RED TIER • STRICT TABOO

Severe Liability Risk

Scenarios:

- Legal evidence and courtroom documentation.
- Medical imaging diagnostics.
- Direct-to-print packaging with certified claims.
- Official corporate logo modifications.

AI autonomy strictly forbidden.

Live Demo: From Chaos to Control

Running demo_repo scripts: hero generation, comparing edits, and programmatic prompt assembly

ROUND 1 • 01_GENERATE.PY

First Call & Base64

- API request to `gemini-3.1-flash-image` .
- Response format: `aspect_ratio: '16:9'` .
- Base64 byte decoding via Python.
- Persisting asset to disk in `out/` .

ROUND 2 • 02_EDIT.PY

Vague vs Controlled

- `--mode vague` : naive prompt breaks handle curvature and glaze.
- `--mode controlled` : sacred constraints preserve 100% of mug geometry while swapping background to travertine.

ROUND 3 • 03_BUILD_PROMPT.PY

JSON Schema Assembly

- Reading schema from `visual_spec.json` .
- Deterministic assembly of Subject, Camera, Constraints, and Ratio.
- Eliminating prompt injection and human typos.

Decision Matrix: GenAI vs Classical CV vs Deterministic Tools

Problem first, not AI first: identifying when generative diffusion is unnecessary and harmful

SOFTWARE 1.0 • DETERMINISTIC

Figma / CSS / SVG / Pillow

When to choose:

- Vector logos and trademarks.
- Crisp typography, pricing, dates.
- Grid-aligned banner composition.

\$0 cost, 0ms latency, 100% typographic accuracy.

SOFTWARE 2.0 • CLASSICAL CV

OpenCV / YOLO / Canny

When to choose:

- Conveyor belt defect detection.
- Barcode and license plate recognition.
- Face cropping and edge alignment.

2-5ms local CPU inference, \$0 API tokens.

SOFTWARE 3.0 • GENERATIVE AI

Diffusion & Multimodal APIs

When to choose:

- E-commerce background replacement.
- Photorealistic texture and lighting synthesis.
- Creative asset variations.

Semantic reasoning over high-dimensional pixels.

Homework: Reproducible Image Workflow Pack

Engineering a reproducible generation package with formal brief and quality scorecard

CHOOSE ONE SCENARIO

3 Real-World Tracks

- 1. **E-commerce:** Product background replacement preserving geometry (ceramics, watch, electronics).
- 2. **YouTube / Media:** Video thumbnail plate with clean negative space for typography (Creator Tools benchmark).
- 3. **Tech Event:** Conference announcement banner with presentation-safe zones.

ARTIFACT CHECKLIST • GENAI-HOMEWORKS/L04/

- ✓ `brief.md` — completed Visual Generation Canvas (goals, constraints, criteria).
 - ✓ `references/` — source input assets (no unreleased private data).
 - ✓ `prompt_v1.txt` & `prompt_v2.txt` — prompt iteration audit trail.
 - ✓ `candidates/` — minimum 2 candidate generation outputs.
 - ✓ `qa.md` — defect audit scorecard evaluating failure modes.
 - ✓ `selected/` — final accepted asset with recorded execution metadata.
- Branch `lesson-04` , PR to `main` , invite: `ihar_rubanovich@epam.com` .

Wrap-up, Open Mic & Next Session Teaser

3 core engineering takeaways and previewing our transition into video generation

KEY TAKEAWAYS

- 1. **Production imagery is a specification, not roulette.** Pinning geometry and sacred constraints outweighs decorative adjectives.
- 2. **Masks and reference signals provide hardware-level control.** Decouple style, identity, and layout conditioning.
- 3. **The 90% iceberg governs production reliability.** Object storage, Base64 decoding, and review gates determine success.

NEXT SESSION • LESSON 05

Video Generation Models & APIs

Continuing **Module 2: Multimodality**.

Expanding from static pixels into the temporal domain: **Runway Gen-3, Kling, OpenAI Sora, Pika**. Camera Motion Control, frame interpolation, and end-to-end video pipelines.

 Microphones open — questions welcome in voice and chat!